

Hemanth Reddy Samidi

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Summary

- Full-stack Engineer with over 3 years of experience, proficient in Angular, React, Node.js, ASP.NET Core, Flask, and Docker. Proven expertise in developing scalable web applications, optimizing data workflows, and enhancing performance and security.
- Demonstrated leadership in cross-functional teams, effectively managing timelines and delivering high-quality solutions.
- **Achievements:** Received the Excellent Execution Award at Micron Technology for enhancing team productivity and meeting key milestones. Granted a Full Scholarship from UMBC with a Data Structures Teaching Assistant position.

Skills

- **Languages & Databases:** Python, C#, TypeScript, JavaScript, Java, C++, MySQL, MSSQL, PostgreSQL, Oracle, NoSQL, Redis
- **Frameworks & Libraries:** Angular, React, Vue.js, Node.js, NgRx, Redux, Next.js, Express.js, npm, Jasmine, Karma, Jest, jQuery, Bootstrap, Material-UI, RxJS, Ag-Grid, .NET Core, ASP.NET Core, Blazor, xUnit, AutoMapper, Flask, Django, FastAPI, Pytest
- **DevOps and Others:** Git, Bitbucket, GitLab, SVN, Docker, Kubernetes, Jenkins, TeamCity, Octopus Deploy, Azure DevOps, Terraform, YAML, Webpack, Artifactory, JIRA, Confluence, Visual Studio, PyCharm, Postman, Swagger, GraphQL, Selenium, TDD, Agile methodology, Scrum, SDLC, SOLID, Microservices, Design patterns, MVC, Windows, Linux, JSON, ETL, Web Services
- **Web Development:** HTML, CSS, RESTful APIs, Responsive Design, Developer Tools, Performance Optimization, UX/UI Design
- **Data Visualization:** Tableau, Power BI, D3.js, Google Chart API, Plotly, Microsoft Excel, Highcharts, Matplotlib, Seaborn
- **Machine Learning:** Regression, Classification, Statistics, Computer Vision, Data Analytics, NLP, Generative AI, LLM, PyTorch
- **Cloud Services:** AWS (Dynamo DB, RDS, Lambda), Azure (Blob Storage, Databricks, App Service, CosmosDB), Google Cloud

Work Experience

- **Software Development Researcher** January 2023 – present
 - DAMS Lab, University of Maryland Baltimore County
 - **IoT Data Collection:** Engineered a framework for discovering, collecting, and analyzing smart device privacy policies. Developed web scrapers in Python, handling over 50,000 records regularly, utilizing NLP and ML algorithms for user insights.
 - **Automated Policy Extraction:** Created a library using Boilerpy3 for automated extraction and custom models for analysis of privacy policies, increasing processing efficiency by 36%. Streamlined data preprocessing workflows, enhancing overall reliability.
 - **Data Analysis and Visualization:** Performed topic modeling and semantic analysis on 956 privacy policies. Built an interactive web [application](#) using Flask and React for result visualization, significantly enhancing user understanding and engagement.
 - **Benchmark:** Pioneered a benchmark suite for evaluating GenAI in privacy interpretation. Leveraged OpenAI API and proposed metrics for data collection and analysis. Conducted comparative studies on 3 leading systems ChatGPT-4, Copilot, and Gemini.
- **Software Engineer - IT Full Stack Developer** July 2020 – July 2022
 - Micron Technology
 - **Full Stack Architecture:** Developed end-to-end web applications using Angular and ASP.NET Core Web API with a C# backend, connected to MSSQL and Azure DB, enhancing R&D workflows for 500+ users. Incorporated testing with Jasmine and Karma.
 - **API Development:** Architected RESTful APIs in ASP.NET Core. Utilized Entity Framework core for database integration, enhancing data consistency and reliability. Optimized query handling and caching, improving API response times by 30%.
 - **Application Security:** Implemented best security practices, including authentication and authorization using OAuth and JWT.
 - **CI/CD:** Orchestrated continuous delivery using TeamCity and Octopus Deploy. Configured TeamCity to automate builds from Git commits, with Octopus deploying across environments, boosting deployment efficiency by 25% and ensuring consistency.
 - **Docker:** Containerized applications using Docker, reducing setup time by 40% and ensuring consistent deployment across servers.
 - **Database Management:** Managed databases including MSSQL, MongoDB, and Azure SQL Database. Executed complex queries to enhance data retrieval speed and performed data migration by transferring over 2 million records between databases.
 - **Legacy System Upgrade:** Refactored legacy applications, leveraging Angular and .NET, achieving a 35% performance boost and extending the system's operational life. Integrated new features to better align with business needs and improve user experience.

Projects

- **VisionPlatePro:** Designed an ALPR system using YOLOv4 and OCR for traffic monitoring. Processed 10,000+ license plate images with 96% precision, enabling vehicle recognition. Deployed with Flask, Angular, and Docker for reliability across environments.
- **AI-driven Privacy Policy Analysis:** Engineered an ML solution with BERT to monitor privacy policy changes, achieving 80.5% Pearson correlation. Built an Angular [website](#) with Flask backend and MongoDB for data management, enhancing policy analysis.
- **Spam Analyzer:** Developed a React web app to detect bots using an ML model, analyzing 15,000+ tweets with 85% accuracy. Used D3.js, Google Chart API, and Plotly for interactive maps, highlighting spam patterns and ensuring cross-device compatibility.

Publications & Research

- **GenAIPABench: A Benchmark for Generative AI-based Privacy Assistants.** *Proceedings on Privacy Enhancing Technologies Symposium (PETS 2024, an A ranked conference)*, July 2024, vol. 2024, no. 3, pp. 336–352. [DOI](#)
- **A Study of the Landscape of Privacy Policies of Smart Devices** *Preprint*, December 2023. [e-print](#)
- **PrivacyLens: A Framework to Collect and Analyze the Landscape of Past, Present, and Future Smart Device Privacy Policies.** *Preprint*, August 2023. [e-print](#).

Education

- **University of Maryland Baltimore County** Maryland, USA
 - Master of Science in Computer Science, GPA: **3.8/4** August 2022 – May 2024
- **Osmania University** Hyderabad, India
 - Bachelor of Engineering in Computer Science, GPA: **3.6/4** August 2016 – September 2020